AICTE APAAR Platform

Student Data Management & Verification System

SIH 2026 · Problem Statement 26263
AICTE Regulation Bureau
Theme: Smart Automation
Repo: *savy-032007/SIH-2026*

Executive Purpose: The AICTE APAAR Platform establishes a single, unified student-level data submission and real-time automated verification gateway keyed by the 12-digit **APAAR ID** (Automated Permanent Academic Account Registry / ABC ID). It resolves the national problem of inconsistent reporting across technical institutions, enforces AICTE Approval Process Handbook intake caps, prevents ghost student enrollments, and integrates with the official **API Setu / DigiLocker Sandbox** for instant academic credential verification.

1. Repository Structure & Artifacts Overview

The savy-032007/SIH-2026 repository contains a production-ready, full-stack microservice and web architecture:

File / Directory	Component	Functional Responsibility
index.html	Institution Portal UI	Complete HTML5 institutional portal interface covering Dashboard, Student Records, Add/Edit Student form, Validation Console, Reports, and Settings.
style.css	Design System	Lightweight, responsive AICTE design system (no external CSS dependencies), styled KPI cards, custom modal, SVG chart layouts, and data tables.
app.js	Frontend Controller	Modular ES6 client engine managing JWT authentication, API Setu verification, dynamic SVG Enrolment Trend curves, category donut charts, and CSV exports.
services/api/	FastAPI Core Backend	REST API engine powered by FastAPI (Python 3.12+), SQLAlchemy 2 sync ORM, Alembic migrations, Pydantic v2 schemas, and security modules.
services/api/app/integrations/	Government Adapters	**ApiSetuSandboxDigiLocker** (official API Setu ABC ID card endpoint) & MockDigiLocker, plus **AbcRegistry** adapter with retry/backoff.
services/api/app/routers/	API Routers	Modular endpoint controllers: institution.py (dashboard, students, verify), auth.py (MFA/TOTP), audit_log.py (Merkle verification).
apps/web/	Next.js 15 Web Portal	Alternative modern App Router application with TypeScript, Tailwind CSS, shadcn/ui, Recharts, and multi-language i18n support.
services/mock-abc/	Mock ABC Registry	Standalone mock service running on port 9001 injecting configurable chaos, latency, and 503 retries for offline simulation.
data/generator/	Synthetic Data Generator	Deterministic generator using Faker('en_IN') producing ~35,000 synthetic students, 100 colleges, and planted fraud anomalies for demo verification.
docker-compose.yml	DevOps Orchestration	Multi-container composition orchestrating PostgreSQL 16, FastAPI Backend (8000), Mock ABC (9001), and Next.js Web (3000).
SPEC.md	Formal Specification	374-line comprehensive regulatory specification covering requirements R1 to R10, data models, and V1 to V5 rule catalogs.

2. Complete Technology Stack & Topology

The platform is engineered around high performance, mathematical determinism, and Zero-Trust government compliance:

Layer	Core Technologies	Architectural Rationale & Capabilities
Frontend	HTML5, CSS3, ES6 JavaScript Next.js 15, TypeScript, Tailwind	Zero external script dependencies on the static portal; instant load times; native SVG vector charts; dynamic conic-gradient donut graphics; full mobile responsiveness.
Backend API	FastAPI, Python 3.12+, Uvicorn, Pydantic v2	Thread-pooled synchronous execution keeping database advisory locks deterministic; OpenAPI auto-documentation; strict typing and declarative request parsing.
Database	PostgreSQL 16, psycopg3, SQLAlchemy 2, Alembic	ACID compliant transactional storage, versioned snapshot history for student enrolments (enrolment_version), and zero-loss migrations.
Cryptography	AES-256-GCM, HMAC-SHA256, Argon2id, PyOTP	Envelope encryption for sensitive PII (APAAR, phone, email, passport); blind deterministic hashes for indexed queries; TOTP MFA for admins; Argon2id password hashing.

Audit Ledger	SHA-256 Cryptographic Hash Chain	Tamper-evident Merkle-style block chain where every record write or PII reveal links to the previous block hash; verified via GET /audit/verify.
Integrations	API Setu Sandbox (DigiLocker), HTTPX, Tenacity	Direct HTTP client querying official endpoint POST /certificate/v3/abc/abcid with HMAC API keys, structured consent, exponential backoff, and local fallback.
Validation & AI	rapidfuzz, scikit-learn, networkx	Token Sort Ratio fuzzy name matching (transliteration invariant); IsolationForest for outlier intake anomalies; NetworkX for ghost student cluster graph analysis.

3. Core System Features & Operational Walkthrough

The platform provides end-to-end capabilities addressing all requirements of SIH Problem Statement 26263:

3.1 Executive Dashboard & Interactive Visualizations

• **Executive KPI Counters:** Displays real-time counts for **Approved Intake** (600), **Actual Enrolment** (570+), **Remaining Seats**, and **Total Students**, dynamically recalculated on student addition or deletion.
• **Dynamic SVG Enrolment Trend Line Chart:** A vector graphic showing actual enrolment from **2022 to 2026** against the 600-seat approved intake ceiling. Features smooth cubic Bezier splines, a soft gradient fill, circular markers, hover tooltips, and a callout badge on the active academic year.
• **Category Distribution Donut:** Renders active demographic splits (General, OBC, SC/ST, NRI/Other) using a dynamic CSS conic-gradient ring.
• **How to Use:** Navigate to **Dashboard** in the sidebar. Switch academic years (2026/2025/2024) via the topbar dropdown to inspect historical intake.

3.2 Student Records Lifecycle Management (CRUD)

• **Multi-Filter & Instant Search:** Search across student name, college ID, or 12-digit APAAR ID with instant debounced filtering. Filter simultaneously by Course (B.Tech CSE, AIML, ECE, Mechanical, B.Pharm, MBA), Category, and Verification Status.
• **Add & Edit Student Form:** Capture student personal details (Name, APAAR ID, DOB, Gender), academic details (Course, Admission Year, College ID), and statutory quota flags (Category, Supernumerary Seats, NRI, OCI, and Foreign National status).
• **Action Modal Confirmations:** Editing opens pre-filled forms; deletion triggers modal confirmation and updates audit logs.
• **How to Use:** Click **Student Records** → ■ **Add Student**. Fill out required fields and click **Save Student**.

3.3 Government API Setu / DigiLocker APAAR Verification

• **Government Endpoint Integration:** Directly interfaces with POST https://sandbox.api-setu.in/certificate/v3/abc/abcid using configured API Setu Client IDs and API Keys.
• **Consent Artifact Flow:** Packages structured JSON consent artifacts containing timestamp, consumer ID, data provider, and digital signature algorithm.
• **Cryptographic Audit Hash:** Computes SHA-256 hash of the verified certificate bytes for proof of document authenticity.
• **Resilient Fallback:** Automatically degrades to local format and registry verification if the external government sandbox times out.
• **How to Use:** In the student form, enter a 12-digit APAAR number (e.g. 918273645012) and click **Verify APAAR**. Observe the confirmation message with the SHA-256 document hash.

3.4 Automated Multi-Stage Validation Engine (Rule Groups V1–V5)

The platform executes an automated compliance pipeline that catches errors before data submission:

Rule Group	Severity	Regulatory Check Description
V1: Duplicate Records	ERROR	Prevents identical APAAR IDs from being submitted twice in a batch or twice within the same institution.
V2: APAAR & Name Matching	ERROR / WARN	Enforces 12-digit format, verifies existence in ABC registry, and executes RapidFuzz token-sort name matching (warns if 80-90%, blocks below 80%).
V3: Intake Cap Enforcement	ERROR	Enforces course-level limits: blocks total students > Approved Intake; blocks NRI > 5% cap; enforces 15%/25% supernumerary quota ceilings.
V4: Data Inconsistencies	WARN / ERROR	Flags Foreign Nationals declaring Indian qualifying boards; checks age plausibility (<15 or >60); demands sponsor proof for NRI category.
V5: Cross-Institution Duplicate	CRITICAL	Detects if the same APAAR ID is actively enrolled at two different institutions in the same academic year.

3.5 Tamper-Evident SHA-256 Audit Chain

• **Cryptographic Merkle Chaining:** Every student insertion, update, deletion, or PII reveal appends a block: entry_hash = SHA256(prev_hash || timestamp || actor_id || action || payload).
• **Tamper Verification:** Accessing GET /api/v1/audit/verify walks the chain from the genesis block. If any database administrator or unauthorized party modifies a row directly in PostgreSQL, the hash chain breaks and identifies the exact tampered row number.
• **How to Use:** Navigate to **Settings** → **Security**. Click **Verify Chain** → to execute instant cryptographic chain verification.

3.6 Regulatory Reports & CSV Data Exporter

• **Instant CSV Exporting:** Generates pre-formatted regulatory spreadsheets directly in the browser with one click.
• **Available Reports:** 1) *Student Records Report* (full roster with APAAR status), 2) *Validation Issues Report* (rule codes V1-V5 and anomalies), and 3) *Enrolment Summary Report* (programme capacity vs actual admissions).
• **How to Use:** Navigate to **Reports** in the sidebar. Click ■ **Export CSV** under any card to instantly download the file.

4. API Setu / DigiLocker Technical Specification

The platform integrates with the confirmed DigiLocker ABC ID Card v3 specification:

Endpoint: POST https://sandbox.api-setu.in/certificate/v3/abc/abcid
Request Headers:
• Content-Type: application/json
• X-APISETU-APIKEY: demokey123456ABCD789 (configurable via DIGILOCKER_API_KEY)
• X-APISETU-CLIENTID: in.gov.sandbox (configurable via DIGILOCKER_CLIENT_ID)

Request Payload Structure:

```
{ "txnId": "f7f1469c-29b0-4325-9dfc-c567200a70f7", "format": "pdf", "certificateParameters": { "IdentityType": "NA", "IdentityValue": "123456", "level1": "1", "AdmissionYear": "2021", "UID": "123412341234", // 12-digit APAAR ID "FullName": "Demo user", "DOB": "31-12-1980", // DD-MM-YYYY format "GENDER": "Male" }, "consentArtifact": { "consent": { "consentId": "ea9c43aa-7f5a-4bf3-a0be-e1caa24737ba", "timestamp": "2024-12-03T12:15:26.958Z", "dataConsumer": { "id": "in.gov.sandbox" }, "dataProvider": { "id": "in.gov.abc" }, "purpose": { "description": "APAAR ABC ID Card Verification" }, "user": { "idType": "string", "idNumber": "string", "mobile": "9988776655", "email": "test@email.com" }, "permission": { "access": "string", "dateRange": { "from": "2024-12-03T12:15:26.958Z", "to": "2024-12-03T12:15:26.958Z" } } }, "signature": { "signature": "SHA256withRSA" } } }
```

5. Backend REST API Directory

FastAPI routes exposed under /api/v1:

Endpoint	Method	Auth / RBAC	Purpose & Response
/api/v1/auth/login	POST	Public	Authenticates credentials; issues access/refresh tokens or demands TOTP MFA.
/api/v1/auth/mfa/verify	POST	Public (MFA token)	Verifies 6-digit TOTP code; issues JWT session tokens.
/api/v1/institution/dashboard	GET	INST_ADMIN, OPERATOR	Returns KPI stats, 2022-2026 enrolment trend points, and category distribution.
/api/v1/institution/students	GET	INST_ADMIN, OPERATOR	Paginated student roster with search query and category/status filters.
/api/v1/institution/students	POST	INST_ADMIN, OPERATOR	Creates student record, generates audit log, and runs V1-V5 validation checks.
/api/v1/institution/students/{id}	PUT / DELETE	INST_ADMIN	Updates or removes student record; snapshots previous version in audit trail.
/api/v1/institution/students/verify-apaar	POST	INST_ADMIN, OPERATOR	Queries API Setu Sandbox ABC ID endpoint; returns verified status and SHA-256 hash.
/api/v1/institution/validation	GET	INST_ADMIN, OPERATOR	Returns summary metrics and detailed breakdown of active validation anomalies.
/api/v1/institution/profile	GET / PUT	INST_ADMIN	Inspects or updates institution AICTE approval number, intake, and address.
/api/v1/audit/verify	GET	INST_ADMIN, AICTE_*	Walks Merkle hash chain from genesis to tail; flags any direct DB tampering.
/api/v1/rules	GET	Authenticated	Returns active AICTE approval process rule parameters for current academic year.

6. Operational Deployment & Execution Manual

Commands to execute, test, and demonstrate the platform:

A. Running with Docker Compose (Multi-Container Deployment):

```
docker compose up --build
```

• Web UI: http://localhost:3000 | FastAPI Backend: http://localhost:8000

• Interactive Swagger Docs: http://localhost:8000/docs | Mock ABC Registry: http://localhost:9001

B. Running Locally with Python Virtual Environment:

1. Activate virtual environment: `.\.venv\Scripts\activate`
2. Start Backend: `$env:PYTHONPATH="services/api"; uvicorn app.main:app --port 8000 --reload --app-dir services/api`
3. Open Frontend: Open index.html directly in any web browser.
4. Default Demo Login: **admin@xyzcollege.ac.in** / **Demo@1234**

C. Running Automated Unit & Integration Tests:

```
$env:PYTHONPATH="services/api"; .\.venv\Scripts\pytest services/api/tests -v
```

• Validates all 5 test suites including API Setu live contract validation, Mock consent, and endpoint checks.